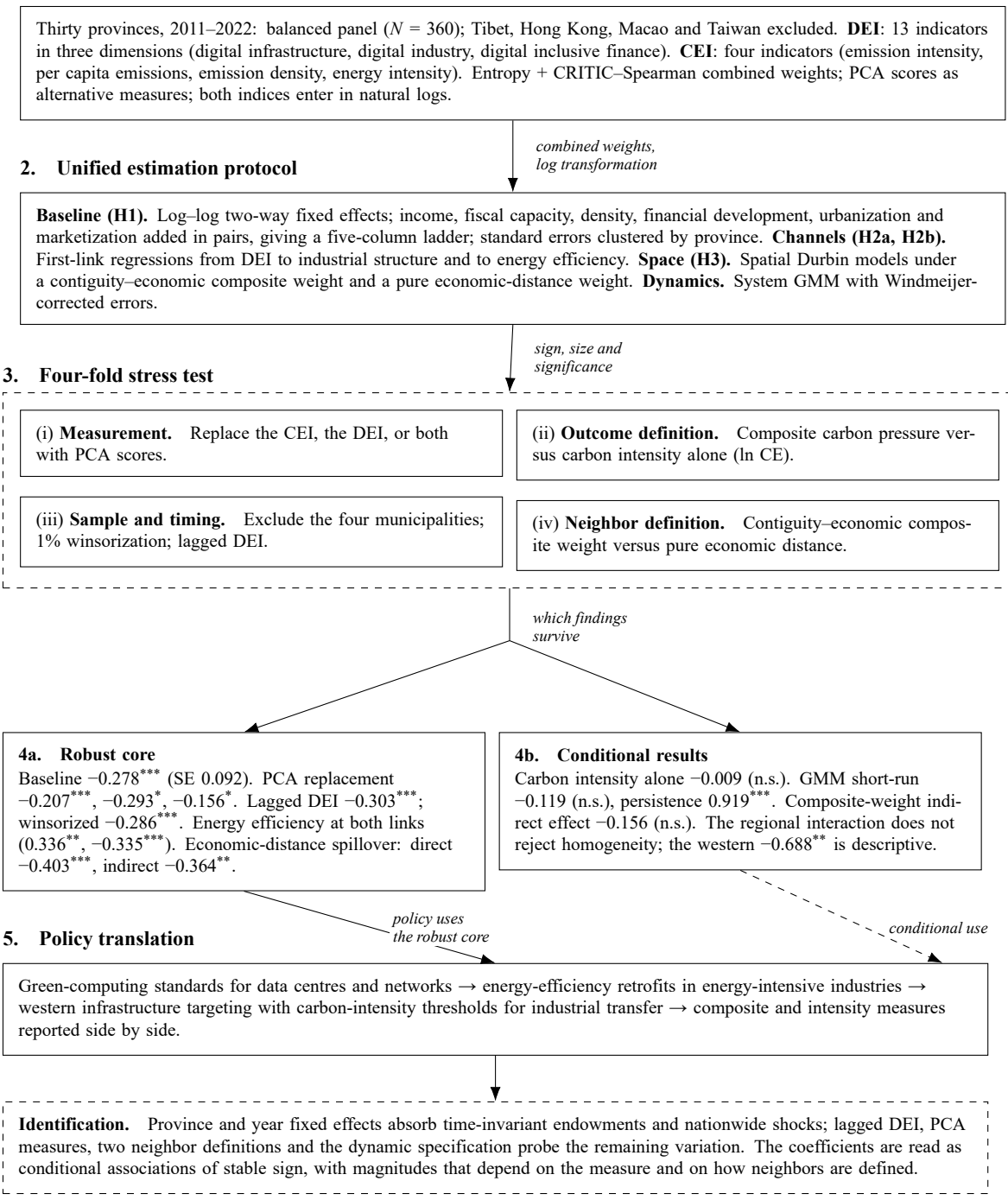

1. Data and construct mapping
Thirty provinces, 2011–2022: balanced panel (N = 360); Tibet, Hong Kong, Macao and Taiwan excluded. DEI: 13 indicators in three dimensions (digital infrastructure, digital industry, digital inclusive finance). CEI: four indicators (emission intensity, per capita emissions, emission density, energy intensity). Entropy + CRITIC–Spearman combined weights; PCA scores as alternative measures; both indices enter in natural logs.
combined weights, log transformation
2. Unified estimation protocol
Baseline (H1). Log–log two-way fixed effects; income, fiscal capacity, density, financial development, urbanization and marketization added in pairs, giving a five-column ladder; standard errors clustered by province. Channels (H2a, H2b). First-link regressions from DEI to industrial structure and to energy efficiency. Space (H3). Spatial Durbin models under a contiguity–economic composite weight and a pure economic-distance weight. Dynamics. System GMM with Windmeijer-corrected errors.
sign, size and significance
3. Four-fold stress test
(i) Measurement. Replace the CEI, the DEI, or both with PCA scores.
(ii) Outcome definition. Composite carbon pressure versus carbon intensity alone (ln CE).
(iii) Sample and timing. Exclude the four municipalities; 1% winsorization; lagged DEI.
(iv) Neighbor definition. Contiguity–economic composite weight versus pure economic distance.
which findings survive
4a. Robust core
Baseline −0.278*** (SE 0.092). PCA replacement −0.207***, −0.293*, −0.156*. Lagged DEI −0.303***; winsorized −0.286***. Energy efficiency at both links (0.336**, −0.335***). Economic-distance spillover: direct −0.403***, indirect −0.364**.
4b. Conditional results
Carbon intensity alone −0.009 (n.s.). GMM short-run −0.119 (n.s.), persistence 0.919***. Composite-weight indirect effect −0.156 (n.s.). The regional interaction does not reject homogeneity; the western −0.688** is descriptive.
policy uses the robust core
conditional use
5. Policy translation
Green-computing standards for data centres and networks → energy-efficiency retrofits in energy-intensive industries → western infrastructure targeting with carbon-intensity thresholds for industrial transfer → composite and intensity measures reported side by side.
Identification. Province and year fixed effects absorb time-invariant endowments and nationwide shocks; lagged DEI, PCA measures, two neighbor definitions and the dynamic specification probe the remaining variation. The coefficients are read as conditional associations of stable sign, with magnitudes that depend on the measure and on how neighbors are defined.